

Physics 4700H: Assignment IV

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Problem 1. Consider two systems of distinguishable noninteracting spin-1/2 particles in a uniform magnetic field **B**. Each spin has a magnetic moment along the field direction $\mu = \gamma\hbar m$ where $m = \pm 1/2$. The energy, U , of a system is proportional to its spin excess $X \equiv N_{\uparrow} - N_{\downarrow}$.

$$U = - \sum_{i=1}^N \boldsymbol{\mu}_i \cdot \mathbf{B} = -(N_{\uparrow} - N_{\downarrow}) \frac{\gamma\hbar}{2} B = -X \frac{\gamma\hbar}{2} B.$$

System 1: $N_1 = 8$ $X_{1i} = 4$

System 2: $N_2 = 4$ $X_{2i} = -2$ Before the systems are put in thermal contact

- (a) What are the initial multiplicity of System 1 (Ω_{1i}), the initial multiplicity of System 2 (Ω_{2i}), and the initial multiplicity of the two systems together (Ω_i)?
- (b) What is an accessible microstate of this combined system before thermal contact?

After the systems are put in thermal contact

- (c) What are N , $N_{\uparrow}$, and $N_{\downarrow}$ for the combined system?
- (d) What is the spin excess, X , of the combined system?
- (e) What is a microstate of the combined system which is only accessible after thermal contact?
- (f) Calculate the multiplicity of the combined system, Ω_f , using the binomial expansion.
- (g) Calculate the multiplicity of the combined system, Ω_f , using

$$\Omega(N, X) = \sum_{X_1} \Omega_1(N_1, X_1) \Omega_2(N_2, X - X_1).$$

- (h) What is the most probable value of X_1 in the combined system ($X_{1\max}$)?

Solution 1.

- (a) For a binary system such as this one

$$\Omega(N, X) = \frac{N!}{(\frac{1}{2}(N+X))! (\frac{1}{2}(N-X))!}$$

and so $\Omega_{1i} = 28$ and $\Omega_{2i} = 4$ and $\Omega_i = \Omega_{1i}\Omega_{2i} = 112$.

- (b) One which preserves the associated X values for systems 1 and 2. E.g. system 1 in $\uparrow\uparrow\uparrow\uparrow\uparrow\downarrow$ and system 2 in $\uparrow\downarrow\downarrow$ and so the whole system is in

$$\uparrow\uparrow\uparrow\uparrow\uparrow\downarrow \cdot \uparrow\downarrow\downarrow.$$

- (c) $N = N_1 + N_2 = 12$, $N_{\uparrow}$ and $N_{\downarrow}$ can be determined by noting that, despite the above microstate having a specific ordering for the spins, any accessible microstate of the same energy would have to have the same number of up and down spins, hence $N_{\uparrow} = 7$ and $N_{\downarrow} = 5$.
- (d) $X = N_{\uparrow} - N_{\downarrow} = 7 - 5 = 2$.

- (e) One which preserves the net $X = 2$ but does not preserve the X values of the individual systems, e.g.

$$\uparrow\uparrow\uparrow\uparrow\downarrow\downarrow\downarrow \cdot \uparrow\uparrow\downarrow\downarrow.$$

- (f) We want to calculate how many ways we can have a system with $X = 2$ for $N = 12$. This is just choosing $N_\uparrow = 7$ spins to be up, and letting the rest be down, i.e.

$$\binom{12}{7} = 792.$$

- (g) As in class I will use a table to determine the allowed values X_1 , i.e. those which give a possible value of $X_2 = X - X_1$

X_1	$X_2 = X - X_1$	Ω_1	Ω_2	$\Omega_f = \Omega_1\Omega_2$	Legal?
8	$2 - 8 = -6$				No
6	-4	8	1	8	Yes
4	-2	28	4	112	Yes
2	0	56	6	336	Yes
0	2	70	4	280	Yes
-2	4	56	1	56	Yes
$\vdots$					No

and since anything with $X_1 < -2$ will continue to increase X_2 past 4, its highest possible value, no other X_1 values are valid and we end the table there. So the “net” multiplicity is the sum of the products, $8 + 112 + 336 + 280 + 56 = 792$, as we got in the previous part.

- (h) The one with the highest multiplicity, given that all states are equally likely, $X_1 = 2$. Like in class, this is not the most likely state for the X_1 system in isolation, but when combined with the X_2 system, sufficient combined states are available as to increase its multiplicity past the $X_1 = 0$ state.

Problem 2. In class, we considered two systems with N_1 and N_2 particles respectively where N_1 and N_2 are large. Initially, they have energies U_{1i} and U_{2i} . After they are brought into thermal contact, $U = U_{1i} + U_{2i}$, and we found that $\Omega_1\Omega_2$ is maximum when $U_1 = U_{1\max} = N_1U/N$. In addition we considered the multiplicity when $U_1 = U_{1\max} + \delta$, and we found that

$$\frac{\Omega(N, U, U_{1\max} + \delta)}{\Omega(N, U, U_{1\max})} = e^{-\alpha\delta^2/N_1} e^{-\alpha\delta^2/N_2} \text{ where } \alpha \equiv \frac{2}{(\gamma\hbar B)^2}.$$

This is not actually a probability but instead just a ratio of two values $\Omega_1\Omega_2$ corresponding to two values of U_1 . In this problem, let's sort out more precisely the probability of finding the system in a state with $U_1 > U_{1\max} \pm$ (that is in a state outside the central peak). As in class, we'll use $N_1 = N_2 = 10^{22}$ and $\delta = 10^{12}/\sqrt{\alpha}$. The probability we want is the ratio of two values: The numerator (NUM) is the sum of the multiplicities for all values of U_1 for which U_1 is either greater than $U_{1\max} + \delta$ or less than $U_{1\max} - \delta$. The denominator (DEN) is the sum of the multiplicities over all possible values of U_1 .

- (a) Find a (symbolic) expression for the denominator starting from

$$DEN \approx \int_{-\infty}^{\infty} \Omega(N, U, U_{1\max}) e^{-\alpha u^2/N_1} e^{-\alpha u^2/N_2} du.$$

- (b) Find a (symbolic) expression for the numerator starting from

$$NUM \approx \int_{\delta}^{\infty} \Omega(N, U, U_{1\max}) e^{-\alpha u^2/N_1} e^{-\alpha u^2/N_2} du.$$

Find a (symbolic) expression for this in terms of the complementary error function:

$$\text{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^{\infty} e^{-t^2} dt.$$

- (c) Now take the ratio NUM/DEN and evaluate using the values of N_1 , N_2 , and δ given above.

Solution 2.

- (a) This is not a hard integral to do. First we combine the two exponentials (call the integral I_1)

$$I_1 = \Omega(N, U, U_{1\max}) \int_{-\infty}^{\infty} e^{-\alpha u^2 \left[\frac{N_1 + N_2}{N_1 N_2} \right]} du$$

and make the substitution $t = \sqrt{\alpha\beta}u \implies dt = \sqrt{\alpha\beta}du$ for $\beta = (N_1 + N_2)/N_1 N_2 = 2 \times 10^{-22}$ which transforms the integral to

$$I_1 = \frac{\Omega(N, U, U_{1\max})}{\sqrt{\alpha\beta}} \int_{-\infty}^{\infty} e^{-t^2} dt$$

where we need not change the bounds as $u = \pm\infty \implies t = \pm\infty$. This is just a standard Gaussian integral and so comes out to $\sqrt{\pi}$, yielding

$$I_1 = \frac{\Omega(N, U, U_{1\max})}{\sqrt{\pi\alpha\beta}}$$

- (b) Call the integral I_2 . We again combine the exponentials and use the same definition for β

$$I_2 = 2\Omega(N, U, U_{1\max}) \int_{\delta}^{\infty} e^{-\alpha\beta u^2} du$$

and we make the same substitution for $t = \sqrt{\alpha\beta}u$ transforming the integral to

$$I_2 = \frac{2\Omega(N, U, U_{1\max})}{\sqrt{\alpha\beta}} \int_{\sqrt{\alpha\beta}\delta}^{\infty} e^{-t^2} dt$$

where here we did need to change the lower bound as δ is finite. Using the given error function identity this integral “evaluates” to

$$I_2 = \frac{2\sqrt{\pi}\Omega(N, U, U_{1\max})}{\sqrt{\alpha\beta}} \operatorname{erfc}(\sqrt{\alpha\beta}\delta).$$

- (c)

$$\begin{aligned} \frac{NUM}{DEN} &\approx \frac{\frac{2\sqrt{\pi}\Omega(N, U, U_{1\max})}{\sqrt{\alpha\beta}} \operatorname{erfc}(\sqrt{\alpha\beta}\delta)}{\frac{\Omega(N, U, U_{1\max})}{\sqrt{\pi\alpha\beta}}} \\ &= \frac{2\sqrt{\pi}\Omega(N, U, U_{1\max}) \operatorname{erfc}(\sqrt{\alpha\beta}\delta) \sqrt{\pi} \sqrt{\alpha\beta}}{\sqrt{\alpha\beta}\Omega(N, U, U_{1\max})} \\ &= 2\pi \operatorname{erfc}(\sqrt{\alpha\beta}\delta) \\ &= 2\pi(1 - \operatorname{erf}(\sqrt{2}10^{-11}10^{12})) = 2\pi(1 - \operatorname{erf}(10\sqrt{2})). \end{aligned}$$

Plugging that into SageMath I obtain

```
[1]: show(N(2*pi*(1-erf(sqrt(2)*10)), prec=512))
```

[1]:

3.460306120704405305148892561702650171704972399133244869217155326584265
25521418956578529376665302960845474399137567655796603836973700352396318
381498018335 $\times 10^{-88}$

The `N` function in Sage gives a numerical approximation to `prec` bits of precision. Ordinary floating point precision, 53 bits for the actual number, isn't enough to distinguish this number from zero.